18000 values have been replaced by 18000 transistors. Now the size of the computer was reduced to 5' (feet). The first transistor based computer is shown in Figure 9.3.

Third Generation Computer (1964 – 1971)

This third generation computer was developed using Integrated Circuits(ICs). An IC is nothing but group of transistors. For example, instead of 100 transistors, if we are using 1 IC, then for 18000 transistors it is enough to use 180 ICs. We know IC is a small device, occupying minimum space. Hence, the size of the computer becomes considerably small. On arrival of third generation, the computer was able to place it on a table. A third generation computer made up of ICs is shown in Figure 9.4.



Figure 9.4 Third Generation Computer

Fourth Generation Computer (1971 of above)

This was developed by using a big IC called Very Large Scale Integrated Circuit (VLSI).The VLSI is nothing but group of ICs. For example, 100 numbers of ICs have been put together and made a single big IC, i.e., VLSI. And this big IC is termed as Microprocessor (µp).The computer what we are using now are belongs to this fourth generation, which is shown in Figure 9.5



Figure 9.5 Fourth Generation Computer

Fifth Generation Computer

Further, the development was going-on but yet to complete. In this, computer scientists want to prove that the computer having some Artificial Intelligence (AI), which is under development stage. The fifth generation computer and its processors as shown in Figure 9.6



Figure 9.6 Fifth Generation Computer

9.3 LANGUAGES



Apart from human, the term *language* is used or needed is only in computer field.

Why the computer need a language?

If a human want to converse with other, certainly he/she is in need of a language. Likewise, if we want to interact with the computer, we are in need of a language.

Generation of Language

Hence, during the development of first generation computer, there was a language

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